

Møller Polarimetry for PV Experiments at 12 GeV

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MOLLER Review

Outline

- 1 Introduction
- 2 Ongoing Upgrade in Hall A
- 3 Møller with Atomic Hydrogen Target
- 4 Conclusion
- 5 Appendix

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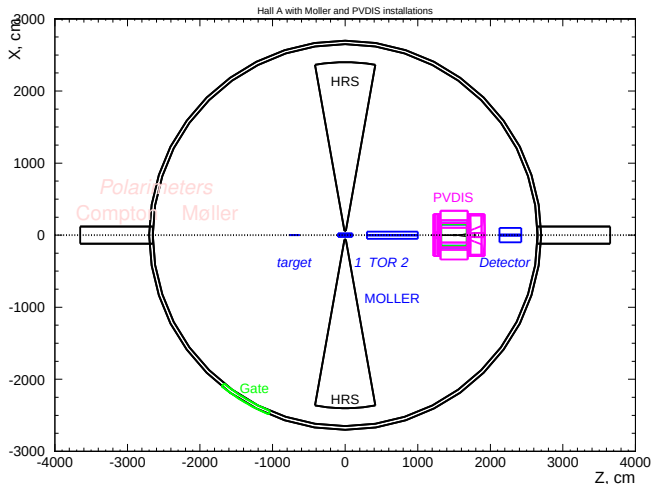
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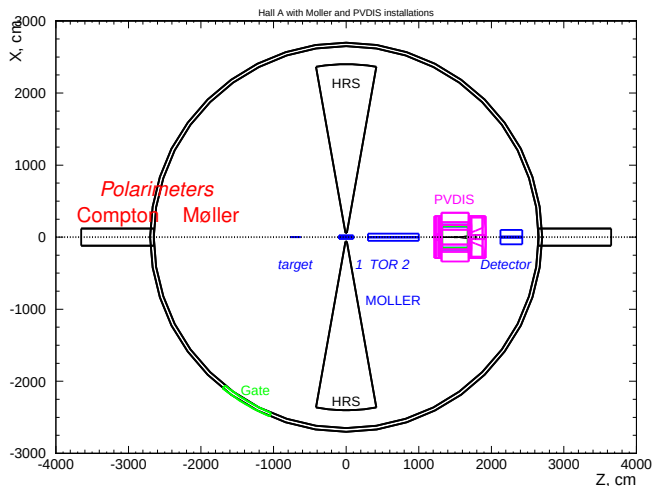
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Installation in Hall A



Installation in Hall A



Error Budget of Møller and SoLID Experiments

Møller

Source of error	% error
Q^2 absolute value	0.5
beam polarization	0.4
beam second order	0.4
inelastic ep	0.4
elastic ep	0.3
other	0.5
total	1.0

SoLID

Source of error	% error
beam polarization	0.4
radiative corrections	0.3
Q^2 absolute value	0.2
statistics	0.3
total	0.6

0.4% - can it be done?

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Electron Polarimetry for PV at JLab: Features

- Energy range $E_{beam} = 6.6 - 11 \text{ GeV}$
- Current range $I_{beam} = 40 - 90 \mu\text{A}$

Additional features to consider

- Time needed to achieve $\sim 0.4\%$ statistical error
- Systematic error:
 - Does polarimetry use the same beam as the experiment (energy, current, location)?
 - Continuous or intermittent (invasive?)
- Two different polarimeters/methods highly desirable

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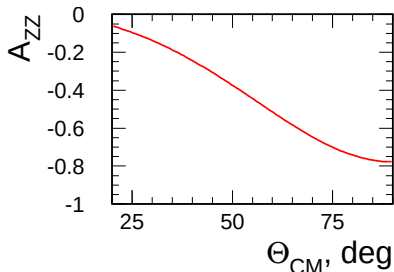
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Møller Polarimetry

- Rad. corrections to Born < 0.3%
- Detecting the e^- at $\theta_{CM} \sim 90^\circ$
- $\frac{dA}{d\theta_{CM}}|_{90^\circ} \sim 0$ - good systematics
- Beam energy independent
- Coincidence - no background
- Ferromagnetic target $\mathcal{P}_T \sim 8\%$
 - Heating $\frac{dP}{dT} \sim 1\%/100^\circ\text{C}$
 $\langle I_B \rangle < 3 \mu\text{A}$
 - Levchuk effect (atomic e^-)
 - Low $\mathcal{P}_T \Rightarrow$ dead time
 - Syst. error $\sigma(\mathcal{P}_T) \sim 2\%$ (0.5%?)

$$\vec{e}^- + \vec{e}^- \rightarrow e^- + e^- \text{ QED.}$$



$$A(E) = -\frac{7}{9}$$

$$\sigma_{lab} \sim 180 \frac{\text{mb}}{\text{ster}}$$

Møller Polarimetry

Polarized electron targets: magnetized ferromagnetic foils

- Iron: polarized d -shell (6 positions occupied out of 10)
- $\mathcal{P}_e$ not calculable: derived from measured magnetization
- Spin-orbital corrections ($\sim 5\%$) - measured in bulk material
- Magnetizing field is along the beam

Field 20 mT, foil at $\sim 20^\circ$

- Magnetization along the foil
- Magnetization can be measured
- A few % from saturation
- Sensitive to annealing, history
- Polarization accuracy $\sim 2 - 3\%$

Field 3 T, foil at $\sim 90^\circ$

- Magnetization perp. to the foil
- Magnetization - from world data
- Foil saturated
- Polarization is robust.
- Polarization accuracy $\sim 0.5\%$

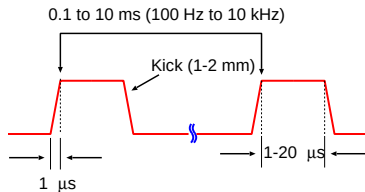
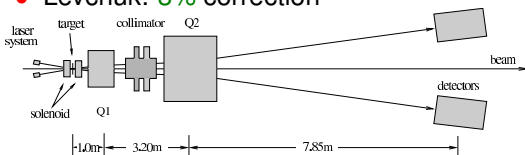
Møller Polarimeter with Saturated Iron foil (Hall C)

JLab, Hall C, M. Hauger *et al.* NIM A **462**, 382 (2001), talk on PAVI09 by S. Page

- External $B_Z \sim 3 - 4 \text{ T}$
- Target foils $1\text{-}10 \mu\text{m}$, perp. to beam
- $\mathcal{P}_t$ not measured
- Levchuk: 3% correction

Tests at high beam current

- Half-moon shape foil
- Kicker magnet

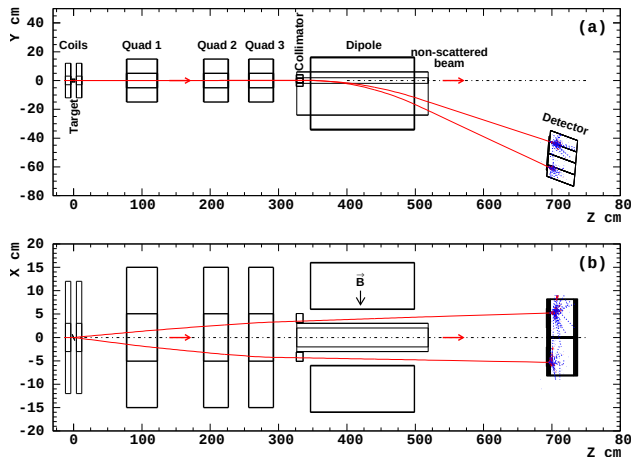


source	$\sigma(A)/A$
optics, geometry	0.20%
target	0.28%
Levchuk effect	0.30%
total at $3 \mu\text{A}$	0.46%
$\Rightarrow 100 \mu\text{A}$	?

A $1 \mu\text{m}$ thick half-foil: mech. problems:

- Foil unstable: holder design
- Thicker foil - high rate
- At $20 \mu\text{A}$ - accidentals/real ≈ 0.4

Hall A Møller Polarimeter



- Minimal Levchuk
- $\sigma_{stat} = 1\%$ in $\sim 2-3$ min
- $B_z \sim 25$ mT field
- Foil at 20° to field
- Foils $5-30 \mu\text{m}$
- Beam $< 2 \mu\text{A}$
- Systematics $\sim 2\%$

Hall A Møller Polarimeter: current upgrade

Upgrade, motivated by PREX requirements:

- High field magnetization (Hall C target clone)
- High instantaneous beam current:
reduce heating by introducing a beam duty cycle $< 5\%$
 - Beam rep. rate $500 \text{ MHz}/4 - \mathcal{F}_{laser} \cdot (n + 1) = \mathcal{F}_{RF} \cdot n$ “beat”
 - “Tune beam”: 4 ms pulses at $\sim 60 \text{ Hz}$
 - Instantaneous counting rate at $50 \mu\text{A}$ will be $\times 3$ higher
 - More invasive than a kicker scheme
- Electronics upgrade to digest higher rates

Target heating with the real raster

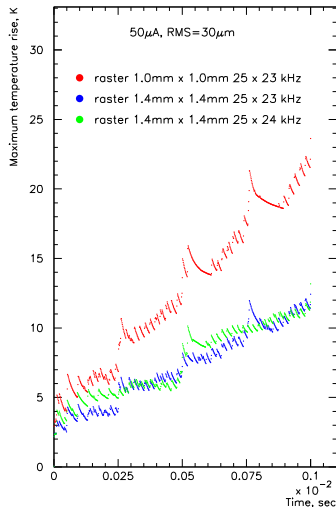
Average heating by $1.5 \mu\text{A}$

- $\Delta T_{max} \sim 22 \text{ K}$ no raster
- $\Delta T_{max} \sim 12 \text{ K}$ raster $1 \times 1 \text{ mm}^2$

Instantaneous heating by $50 \mu\text{A}$

$50 \mu\text{A}$, $\sigma_X \sim 30 \mu\text{m}$, $\Delta t = 1 \text{ ms}$
 Raster $\sim 1.4 \times 1.4 \text{ mm}^2$, $25 \times 24 \text{ kHz}$

- In pulse $\Delta T_{max} \sim 12 \text{ K}$
- Total $\Delta T_{max} \sim 24 \text{ K}$ - acceptable!



Møller Systematic Errors

Variable	Hall C	Hall A	
		Present	Upgrade
Target polarization	0.25%	2.00%	0.50%
Target angle	0.00%	0.50%	0.00%
Analyzing power	0.24%	0.30%	0.30%
Levchuk effect	0.30%	0.20%	0.20%
Target temperature	0.05%	0.00%	0.02%
Dead time	?	0.30%	0.30%
Background	?	0.30%	0.30%
Others	0.10%	0.30%	0.30%
Beam extapolation	?	larger	?
Total	0.47%	2.10%	~0.80%

Possible Breakthrough in Accuracy

Møller polarimetry with 100% polarized atomic hydrogen gas, stored in a ultra-cold magnetic trap.

E.Chudakov and V.Luppov IEEE Trans. on Nucl. Sc., 51, 1533 (2004)

http://www.jlab.org/~gen/hyd/loi_3.pdf

Advantages:

- 100% electron polarization
 - very small error on polarization
 - sufficient rates $\sim \times 0.005$ - no dead time
 - false asymmetries reduced $\sim \times 0.1$
- Hydrogen gas target
 - no Levchuk effect
 - low single arm BG from rad. Mott ($\times 0.1$ of the BG from Fe)
 - high beam currents allowed: continuous measurement

Operation:

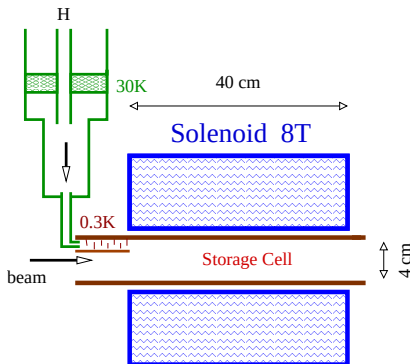
- density: $\sim 6 \cdot 10^{16}$ atoms/cm²
- Stat. error at 50 μ A: 1% in ~ 10 min

Møller Systematic Errors

Proposed: 100%-polarized atomic hydrogen target ($\sim 3 \cdot 10^{16}$ atoms/cm²).

Variable	Hall C	Hall A		
		Present	Upgrade	Proposed
Target polarization	0.25%	2.00%	0.50%	0.01%
Target angle	0.00%	0.50%	0.00%	0.00%
Analyzing power	0.24%	0.30%	0.30%	0.10%
Levchuk effect	0.30%	0.20%	0.20%	0.00%
Target temperature	0.05%	0.00%	0.02%	0.00%
Dead time	-	0.30%	0.30%	0.10%
Background	-	0.30%	0.30%	0.10%
Others	0.10%	0.30%	0.30%	0.30%?
Beam extapolation	?	larger	?	0.00%
Total	0.47%	2.10%	~0.80%	~0.35%

Storage Cell



First: 1980 (I.Silvera,J.Walraven)

$\vec{p}$ jet (Michigan)

Never put in high power beam

- 4 states, $\mathcal{P}_e$: $|a\rangle, |b\rangle = -1$ $|c\rangle, |d\rangle = +1$
- $-\vec{\nabla}(\vec{\mu}_H \vec{B})$ force in the field gradient
 - pulls $|a\rangle, |b\rangle$ into the strong field
 - repels $|c\rangle, |d\rangle$ out of the field
 - $\mathcal{P}_e = 1 - \delta$, $\delta \sim 10^{-5}$
- $H+H \rightarrow H_2$ recombination (+4.5 eV)
high rate at low T
 - parallel electron spins: suppressed
 - gas: 2-body kinematic suppression
 - gas: 3-body density suppression
 - surface: strong unless coated
 ~ 50 nm of superfluid ^4He
- Density $3 \cdot 10^{15} - 3 \cdot 10^{17} \text{ cm}^{-3}$.
- Gas lifetime > 1 h.

Contaminations and Depolarization of the Target Gas

Ideally, the trapped gas polarization is nearly 100% ($\sim 10^{-5}$ contamination).
 Good understanding of the gas properties (without beam).

Contamination and Depolarization

No Beam

Gas Properties

- Atom velocity ≈ 80 m/s
- Atomic collisions $\approx 1.4 \cdot 10^5$ s $^{-1}$
- Mean free path $\lambda \approx 0.6$ mm
- Wall collision time $t_R \approx 2$ ms
- Escape (10cm drift) $t_{es} \approx 1.4$ s

CEBAF Beam

- Bunch length $\sigma = 0.5$ ps
- Repetition rate 497 MHz
- Beam spot diameter ~ 0.2 mm

- Hydrogen molecules $\sim 10^{-5}$
- Upper states $|c\rangle$ and $|d\rangle < 10^{-5}$
- Excited states $< 10^{-5}$
- Helium and residual gas $< 0.1\%$
 - measurable with the beam

100 μ A Beam

- Depolarization by beam RF $< 2 \cdot 10^{-4}$
- Ion, electron contamination $< 10^{-5}$
- Excited states $< 10^{-5}$
- Ionization heating $< 10^{-10}$

Expected depolarization $< 2 \cdot 10^{-4}$

Potential Improvement of Systematic Accuracy

Fe at 3T: potential improvement (quite optimistic):

- Better understanding of magnetization in thin foils (find experts)
- More extensive MC and beam studies
- Measurements: 0.1% stat $\Rightarrow$ 3-5h beamtime $\Rightarrow$ beam stability?

Variable	Hall C	Hall A		
		Fe at 3T	H ₁ gas	
Target polarization	0.25%	0.50%	0.25%	0.01%
Target angle	0.00%	0.00%	0.00%	0.00%
Analyzing power	0.24%	0.30%	0.20%	0.15%
Levchuk effect	0.30%	0.20%	0.20%	0.00%
Target temperature	0.05%	0.02%	0.02%	0.00%
Dead time	-	0.30%	0.15%	0.10%
Background	-	0.30%	0.15%	0.10%
Others	0.10%	0.30%	0.15%	0.15%
Beam extapolation	?	0.15%	0.15%	0.00%
Total	0.47%	0.82%	0.48%	0.25%

Conclusion

New PV experiments require a $\sim 0.4\%$ polarimetry.
Two options for the Møller polarimetry:

Iron foil in strong field

- Not continuous
- Invasive
- Certain: 0.8%
- Potential: 0.5% with R&D
 - manpower ~ 4 FTE*Y
 - low material cost ($\sim 50k$)

Atomic hydrogen

- Continuous
- Not invasive
- Novel instrument
- Potential: 0.25% !
- Possible steps:
 - R&D - cell with electrodes
1 FTE*Y, 50k
- Interest expressed:
 - UVA (Don Crabb)
 - Mainz (Frank Maas)

Bunch suppression

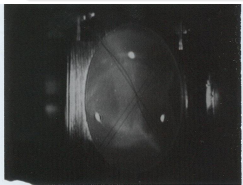
Options (from the draft of a paper by M.Poelker et al)

- G0: laser running at **499/16MHz** - too long to install
- For regular bunch charges: laser at $\mathcal{F}_{laser} < \mathcal{F}_{RF}$
bunch suppression on the chopper.

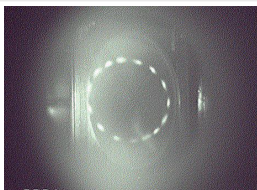
Beat frequency condition ($\mathcal{F}_{RF} = 499\text{MHz}$):

$$\mathcal{F}_{laser} \cdot (n + 1) = \mathcal{F}_{RF} \cdot n,$$

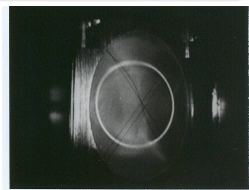
$n = 3, 4, 7, 15, 31, \dots$ - “magic” numbers



regular $\mathcal{F}_{laser} = \mathcal{F}_{RF}$



$n = 15$

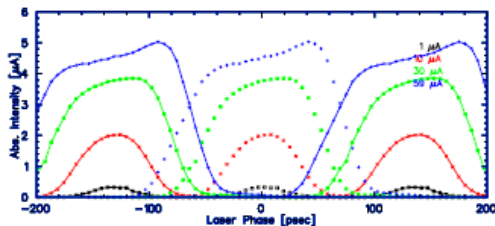


continuous

Beat frequency mode - leak through

Pulses overlap

- $\tau_{pulse} \sim 200$ ps @ $50 \mu A$
- τ_{pulse} grows with $\mathcal{I}_{beam}$ (electro-repulsion)
- Fully open slit 110 ps
- No leak: $\Delta\tau > 160$ ps



Optimization

- $n=15$ same slit $\Delta\tau = 133$ ps, contamination $\sim 5\%$ - bad
- $n=7$ same slit $\Delta\tau = 285$ ps, no contamination; other slit $\Delta\tau = 95$ ps leak $\sim 30\%$ - invasive for other halls
- $n=4$ other slit $\Delta\tau = 166$ ps non-invasive?

Hydrogen Atom in Magnetic Field

$H_1: \vec{\mu} \approx \vec{\mu}_e;$

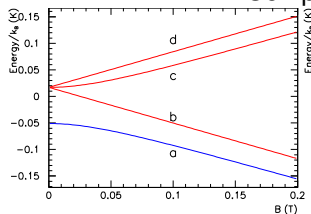
$H_2:$ opposite electron spins

Consider H_1 in $B = 7 \text{ T}$ at $T = 300 \text{ mK}$

At thermodynamical equilibrium:

$$n_+/n_- = \exp(-2\mu B/kT) \approx 10^{-14}$$

Complication from hyperfine splitting:



Low energy

$$|b\rangle = |\downarrow\downarrow\rangle$$

$$|a\rangle = |\downarrow\uparrow\rangle \cdot \cos\theta - |\uparrow\downarrow\rangle \cdot \sin\theta$$

High energy

$$|d\rangle = |\uparrow\uparrow\rangle$$

$$|c\rangle = |\uparrow\downarrow\rangle \cdot \cos\theta + |\downarrow\uparrow\rangle \cdot \sin\theta$$

where $\tan 2\theta \approx 0.05/B(T)$, at 7 T $\sin\theta \approx 0.0035$

Mixture $\sim 53\%$ of $|a\rangle$ and $\sim 47\%$ of $|b\rangle$:

$$\mathcal{P}_e \sim 1 - \delta, \quad \delta \sim 10^{-5},$$

$$\mathcal{P}_p \sim -0.06 \text{ (recombination)} \Rightarrow \sim 80\%$$

Dynamic Equilibrium and Proton Polarization

Proton polarization builds up, because of recombination of states with opposite electron spins:

$$|a\rangle = |\downarrow\uparrow\rangle\alpha + |\uparrow\downarrow\rangle\beta \text{ and}$$

$$|b\rangle = |\downarrow\downarrow\rangle$$

As a result, $|a\rangle$ dies out and only $|b\rangle = \downarrow\downarrow$ is left!

$$P \rightarrow 0.8$$

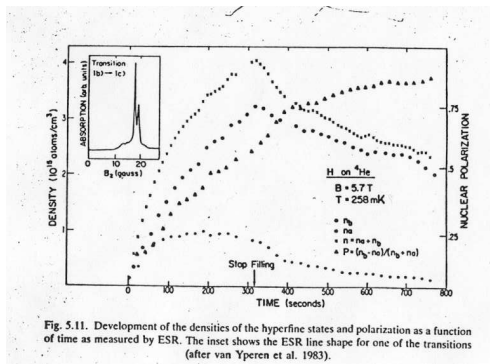


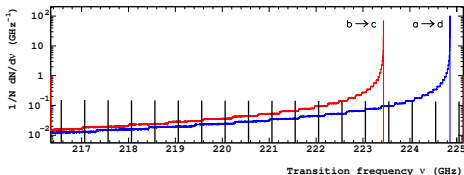
Fig. 5.11. Development of the densities of the hyperfine states and polarization as a function of time as measured by ESR. The inset shows the ESR line shape for one of the transitions (after van Yperen et al. 1983).

Contaminations and Depolarization of the Target Gas

100 μA CEBAF beam:

Beam RF influence

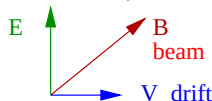
- $|a\rangle \rightarrow |d\rangle$ and $|b\rangle \rightarrow |c\rangle \sim 200 \text{ GHz}$
- RF spectrum: flat at $< 300 \text{ GHz}$



- $\sim 10^{-4} \text{ s}^{-1}$ conversions (all atoms)
- $\sim 6\% \text{ s}^{-1}$ conversions (beam area)
- Diffusion: contamination
 $\sim 1.5 \cdot 10^{-4}$ in the beam area
- Solenoid tune to avoid resonances

Gas Ionization

- 10^{-5} s^{-1} of all atoms
- $20\% \text{ s}^{-1}$ in the beam area
- Problems:
 - No transverse diffusion
 - Recombination suppressed
 - Contamination $\sim 40\%$ in beam
- Solution: electric field $\sim 1 \text{ V/cm}$
 - Drift $\vec{v} = \vec{E} \times \vec{B} / B^2 \sim 12 \text{ m/s}$
 - Cleaning time $\sim 20 \mu\text{s}$
 - Contamination $< 10^{-5}$
 - Ions, electrons: same direction
 - Beam $\vec{E}_r(160 \mu\text{m}) \approx 0.2 \text{ V/cm}$



Summary on Atomic Hydrogen for Møller Polarimetry

Potential for Polarimetry

- Systematic accuracy of $< 0.3\%$
- Continuous measurements
- Tools for systematic studies:
 - changing the electrical field (ionization)
 - changing the magnetic field (RF depolarization)

Problems and Questions

- Electrodes in the cell: R&D is needed
- Residual gas 0.1% accurate subtraction
 - Coordinate detectors: the interaction point?
- Atomic cross section (mean free path...) needs verification
- Cost and complexity